

Azure Backup & Azure Site Recovery (ASR)

Definition: Azure Backup and Azure Site Recovery (ASR) are complementary Microsoft cloud services for business continuity. Azure Backup focuses on long-term data retention and recovery (protecting against data loss and ransomware), while ASR is designed for continuous workload replication and instant failover (minimizing downtime during a disaster).

Azure Backup

Cloud-based backup solution for:

- VMs
- Databases
- Azure Files

Azure Site Recovery (ASR)

Disaster Recovery service replicating workloads to another region/site.

Microsoft Learn:

[Azure Backup Documentation](#)

[Azure Site Recovery Documentation](#)

Use Cases

Backup

- Recover deleted files
- Restore VM after corruption

ASR

- Regional disaster recovery
- Business continuity

Real-Time Scenario

Datacenter Failure

Primary Azure region goes down.

ASR:

- Fails over VMs
- Starts workloads in secondary region
- Business continues running

Step-by-Step: Azure Backup

1. Create Recovery Services Vault
2. Open Vault
3. Click Backup
4. Select workload:
 - Azure VM
5. Configure backup policy
6. Enable backup

Step-by-Step: ASR

1. Open Recovery Services Vault
2. Select Site Recovery
3. Enable replication
4. Select source VM
5. Select target region
6. Configure replication policy
7. Start replication
8. Test failover

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1. Azure Monitor

Definition

Azure Monitor is Microsoft Azure's centralized monitoring platform used to collect, analyze, visualize, and respond to telemetry data from:

- Azure resources
- Virtual Machines
- Applications
- Networks
- Containers
- On-premises servers

It collects:

- Metrics

- Logs
- Traces
- Events

Microsoft Learn:

[Azure Monitor Overview](#)

Use Cases

- Monitor VM CPU & memory usage
- Detect application downtime
- Track network latency
- Monitor Kubernetes clusters
- Create alerts for failures
- Analyze logs for troubleshooting

Real-Time Scenario

Scenario:

An e-commerce website hosted on Azure App Service becomes slow during a festive sale.

Solution Using Azure Monitor:

1. Azure Monitor detects high CPU usage
2. Metrics show CPU spikes
3. Logs reveal failed database queries
4. Alerts notify DevOps team
5. Autoscaling adds more instances

Step-by-Step: Enable Azure Monitor for VM

1. Login to Azure Portal
2. Search for **Virtual Machines**
3. Open VM
4. Go to **Monitoring**
5. Click **Insights**
6. Enable monitoring

7. Install Azure Monitor Agent
8. Connect to Log Analytics Workspace

Azure Monitor Metrics & Azure Monitor Logs

Definition

Metrics

Numerical performance values collected at regular intervals.

Examples:

- CPU %
- Disk IOPS - (Input/Output Operations Per Second) in Azure measures the speed and efficiency of storage devices, representing the number of read/write operations a virtual machine disk or database can handle in one second. Higher IOPS equals better performance, crucial for high-demand applications like SQL databases and, for many disk types, is determined by the Azure VM size and disk type. [[1](#), [2](#), [3](#), [4](#)]
- Memory utilization
- Network throughput

Logs

Detailed event records used for troubleshooting and auditing.

Examples:

- Security events
- Application exceptions
- Failed logins
- System events

Microsoft Learn:

[Azure Monitor Metrics Overview](#)

[Azure Monitor Logs Overview](#)

Key Difference

Metrics	Logs
Lightweight	Detailed
Near real-time	Deep analysis
Numeric values	Text/event records

Metrics	Logs
Faster queries	Complex troubleshooting

Real-Time Scenario

CPU Spike Issue

- Metrics show CPU usage reached 95%
- Logs reveal a scheduled backup job caused overload

Together they help identify:

- What happened
- Why it happened

Step-by-Step: View Metrics

1. Open Azure Portal
2. Open Virtual Machine
3. Select **Metrics**
4. Choose metric:
 - CPU Percentage
 - Disk Read
5. Set time range
6. Analyze graph

Step-by-Step: Query Logs

1. Open Log Analytics Workspace
2. Click **Logs**
3. Run KQL query:

Heartbeat

```
| summarize count() by Computer
```

4. Review results

3. Azure Monitor Alerts

Definition

Alerts notify administrators automatically when a condition occurs.

Alert Types:

- Metric Alerts
- Log Alerts
- Activity Log Alerts
- Smart Detection Alerts

Microsoft Learn:

[Azure Monitor Alerts Overview](#)

Use Cases

- CPU > 90%
- VM stopped unexpectedly
- Application downtime
- Failed login attempts

Real-Time Scenario

Banking Application

If transaction failures exceed 5%:

- Alert triggers
- Email/SMS sent
- Automation restarts service

Step-by-Step: Create Alert

1. Open Azure Monitor
2. Select **Alerts**
3. Click **Create Alert Rule**
4. Select resource
5. Add condition:
 - CPU > 80%
6. Add Action Group

7. Define severity
8. Create alert

4. Azure Monitor Action Groups

Definition

Action Groups define what happens after an alert is triggered.

Actions include:

- Email
- SMS
- Push notification
- Logic Apps
- Azure Functions
- Webhooks

Microsoft Learn:

[Create and Manage Action Groups](#)

Real-Time Scenario

Production VM Down

When VM stops:

- Email sent to admin
- SMS sent to manager
- Azure Function auto-restarts VM

Step-by-Step

1. Azure Monitor → Alerts
2. Open **Action Groups**
3. Create new Action Group
4. Add notification type:
 - Email
 - SMS
5. Add automation action

6. Save

5. Azure Log Analytics Workspace

Definition

Central storage and analysis platform for logs collected from Azure resources.

Uses:

- Store logs
- Run KQL (Kusto Query Language) queries
- Troubleshooting
- Security analysis

Microsoft Learn:

[Log Analytics Workspace Overview](#)

Use Cases

- Centralized monitoring
- Security auditing
- VM health monitoring
- Application diagnostics

Real-Time Scenario

Multi-Region Company

Logs from:

- India servers
- US servers
- Europe servers

All collected into one workspace for analysis.

Step-by-Step: <https://learn.microsoft.com/en-us/azure/azure-monitor/logs/quick-create-workspace?tabs=azure-portal>

Configure a log analytics workspace to collect Window Server Event log, IIS and performance data.

Configuring Azure Monitor with Log Analytics for IIS Servers

Azure Monitor combined with Log Analytics provides centralized telemetry collection for performance metrics, event logs, and application logs from Windows-based workloads. This guide demonstrates how to configure data collection from IIS servers using Data Collection Rules (DCRs).

Create the Log Analytics Workspace

1. Navigate to **Log Analytics workspaces** in the Azure portal
2. Select **Create**
3. Choose your resource group (e.g., Zava IIS resource group)
4. Provide a workspace name and select your preferred region
5. Select **Review + Create**, then **Create**

After deployment, configure RBAC permissions by assigning the Contributor role to users or service principals that need to interact with the workspace data.

Configure Data Collection Infrastructure

Create a Data Collection Endpoint:

1. Navigate to **Azure Monitor** in the portal
2. Select **Data Collection Endpoints**, then **Create**
3. Specify the endpoint name, subscription, resource group, and region (match your Log Analytics workspace region)
4. Create the endpoint

Create a Data Collection Rule:

1. Navigate to **Data Collection Rules** and select **Create**
2. Provide a rule name, resource group, and region
3. Select **Windows** as the platform type
4. Choose the data collection endpoint created in the previous step
5. Skip the **Resources** tab initially (you'll associate VMs later)

Configure Data Sources

Add three data source types to capture comprehensive telemetry:

Performance Counters:

1. On the **Collect and Deliver** page, select **Add data source**
2. Choose **Performance Counters** as the data source type
3. Select **Basic** for standard CPU, memory, disk, and network metrics (or **Custom** for specific counters)
4. Set the destination to **Azure Monitor Logs** and select your Log Analytics workspace

Windows Event Logs:

1. Add another data source and select **Windows Event Logs**
2. Choose **Basic** collection mode
3. Select Application, Security, and System logs
4. Configure severity filters (Critical, Error, Warning for Application and System; Audit Success for Security)
5. Specify the same Log Analytics workspace as the destination

IIS Logs:

1. Add a final data source for **Internet Information Services logs**
2. Accept the default IIS log file paths or customize as needed
3. Set the destination to your Log Analytics workspace

After configuring all data sources, select **Review + Create**, then **Create** the data collection rule.

Associate Resources

1. Navigate to your newly created **Data Collection Rule**
2. Select **Resources** from the rule properties
3. Click **Add** and select your IIS servers (e.g., zava-iis1, zava-iis2)
4. Return to **Data Collection Endpoints**
5. Select your endpoint and add the same IIS servers as resources

This two-step association ensures proper routing of telemetry data.

Query Collected Data

After allowing time for data collection, query the telemetry:

1. Navigate to your **Log Analytics workspace**
2. Select **Logs** to open the query editor
3. Browse **predefined queries** under Virtual Machines
4. Run the "What data has been collected" query to view performance counters, network metrics, and memory data
5. Access **Insights** to monitor data ingestion volumes

You can create custom KQL queries to analyze specific events, performance patterns, or IIS log entries across your monitored infrastructure.

Find out more at: <https://learn.microsoft.com/en-us/azure/azure-monitor/fundamentals/overview>

1. Azure Portal
2. Search **Log Analytics Workspace**
3. Click Create
4. Select:
 - Subscription
 - Resource Group
 - Region
5. Create workspace
6. Connect resources